

Python Data Visualization Assignment Report

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1 Introduction

The aim of this project is to explore data visualization using Python, from selecting the data all the way to telling a story through it's visualization.

1.1 Methodology

The project itself followed the following process to achieve the previously mentioned goal:

1. Choose dataset.
2. Analyze and process data.
3. Think of questions to answer with visualizations.
4. Choose visualizations types.
5. Choose libraries.
6. ...

1.2 Sources

Numerous sources are used through out the project such as:

- AI
- Documentation
- Forums
- ...

Regardless of the source, these are indicated through bibliography/references or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage

A point was made to use AI in this project in a manner in which it's answers are scrutinized and reviewed, to make sure hallucinations are not taken into account and answers are not just implemented without a second thought. This way, and active usage of AI is encouraged with certain restraints in place.

AI being used for this project was also hosted locally, using smaller models to do so making it's usage more reproducible.

2 Dataset

Data for this project was required to not have been selected by a different student. To minimize the risk of this happening, activity data from a Garmin Forerunner 245 was chosen, reducing the chances of another student having the same dataset.

The data set, located in `./data/activities.csv`, consists of:

- 31 columns & 180 rows.
- Data spanning from: 21 February 2021 - 30 March 2026 (5 years).
- Varying data types (integers, floats, dates, time spans, locations, strings).

Having 31 columns allows for many different questions to be asked about the dataset, allowing it to be explored in multiple ways if desired. The limited number of rows (180) whilst still being a decent amount, allows it's content to be analyzed in a more hands on way, being able to be more critical about each specific row and actively engage in processing in a more granular fashion. The dates spanning 5 years back allows for a story to be told through time, and analyze factors like consistency. Lastly, the varying data types allows for different types of visualizations that require different data types to be used such as maps (with location).

2.1 Analysis

Before asking questions about the dataset, the dataset was further analyzed using both the `df.info()` function to check how the data was imported into the pandas dataframe.

A small AI analysis was invoked using `qwen2.5:14b` for the prompt which was the following:

Read the linked CSV file and analyze it, create a table with the following fields:

- N: Column number.
- Column: Column name.
- Description: Content description (what the data in the column usually looks like), excluding null indicators).
- Null indicator: Indicator that seems to be used for when the data is null.
- Data type: Suggested data type to be used when reading CSV in Pandas.

The table generated was then used as a starting point although it's content was scrutinized and interpreted correctly, assessing whether or not the AI had hallucinated when generating it. Some important conclusions reached with the help of this analysis were:

- Some columns weren't really needed or used (such as *Favorite*).
- `-` was used as a null indicator on most columns.
- Correct data types for each column.

2.2 Processing

Processing the data in this case consisted of the following steps:

1. Deleting unused columns.
2. Changing Null indicator for None values when applicable.
3. Clean data (change it to usable format).
4. Set correct data type to column.

These were done using the following libraries:

- Pandas
- NumPy
- re

2.3 Questions

Having analyzed and processed the data, many rows and columns were cleaned up from the dataset and columns were changed to the correct format. Having the data in this format, allowed for questions to be asked since now there is a better idea of the data at hand. The following questions were therefore asked taking into account the current state of the data:

1. What kind of activities are being recorded?
2. Are the recorded activities showing enough weekly activity?
3. Where are activities being recorded, and how do activity patterns differ by location?

Each question and why they were chosen is explained below:

2.3.1 Question 1: Activity Composition

Question 1 is a good starting question because it dives into categorizing data into different types of activity which is a good starting point to understand what type of data is being recorded and later explored in other questions.

2.3.2 Question 2: Weekly Activity Assessment

Question 2 proceeds naturally from question 1 as it benchmarks personal fitness against WHO guidelines (150 min moderate or 75 min vigorous per week). This question requires aggregating Time by week and classifying intensity via Avg HR thresholds, enabling trend analysis over 5 years.

2.3.3 Question ??: Geographic Patterns

Question ?? adds spatial context by extracting locations from Title, mapping activity density and comparing metrics by place. It also incentivizes data to be used in a more creative way, not just looking at the given fields, but extracting a whole different field type to create a new field type such as location in this case from just a title.

3 Data Processing & EDA

4 Visualization Process

5 Research Questions

5.1 Question 1

5.2 Question 2

5.3 Question 3

6 Visualization Summary

7 Visualizations

7.1 Chart 1

Figure 1: Your caption

8 Summary Table

Question	Charts	Chart Type	Library
1	TODO	TODO	TODO
	TODO	TODO	TODO
2	TODO	TODO	TODO
	TODO	TODO	TODO
3	TODO	TODO	TODO

9 Conclusions

AI Usage Declaration

Bibliography